Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [14AE73]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2019

VIBRATIONS OF ELASTIC SYSTEMS

Time:3 Hours **Maximum Marks:70**

Instructions: **i.** Answer any **FIVE** full Questions choosing atleast two from each part.
ii. Missing data may be suitably assumed.

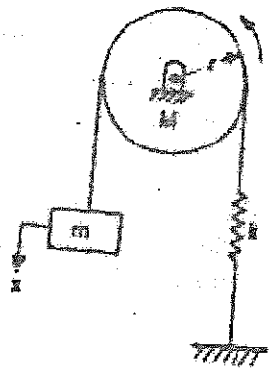
PART - A

1. a. Explain the following types of vibrations **06**
i. Free and forced
ii. Longitudinal, transverse and torsional
iii. Deterministic and random

b. Find the resultant of 2 SHM, $x_1=5\sin 2nt$, $x_2=7\sin(2nt+n/3)$ **08**

2. a. Explain the concept of bending vibration when 2 identical bars are suspended with the help of a string of length a and the bars are connected with each other by a wire of length l and diameter d and are excited to perform torsional oscillations.

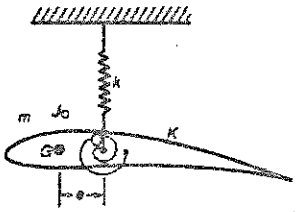
b. Determine the natural frequency of the spring mass pulley system as shown in the figure.



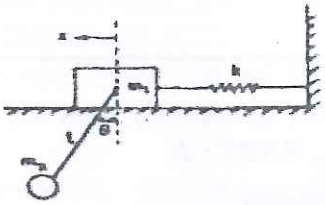
3. a. Obtain the formulations and solutions of a body with a single DOF spring mass damper system under damped free vibrations. **07**

b. The torsional pendulum with a disc of moment of inertia $J=0.05$ kgm^2 immersed in a viscous fluid. During the vibrations of pendulum, the observed amplitudes on the same side of the neutral axis for successive cycles are found to decay 50% of the initial value, Determine, **07**
i. Logarithmic decrement
ii. Damping coefficient of torque/unit velocity
iii. The periodic time of vibration
The frequency when the disc is removed from the fluid.

4. a. Obtain the differential equations of motion and the natural frequencies of vibration of a 2D airfoil supported as shown. M=5 kg, k=5000 N/m, Distance from the leading edge a=0.1 m, J=0.12 kgm^2, K=400 Nm/rad. **05**



b. Find the natural frequencies for the system shown in the figure. K=90N/m, l=0.25, m_1=2kg, m_2=0.5kg. **05**

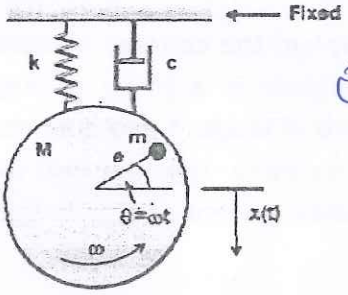


c. Obtain the natural frequencies of a body analysed as a mixed coordinate system. **04**

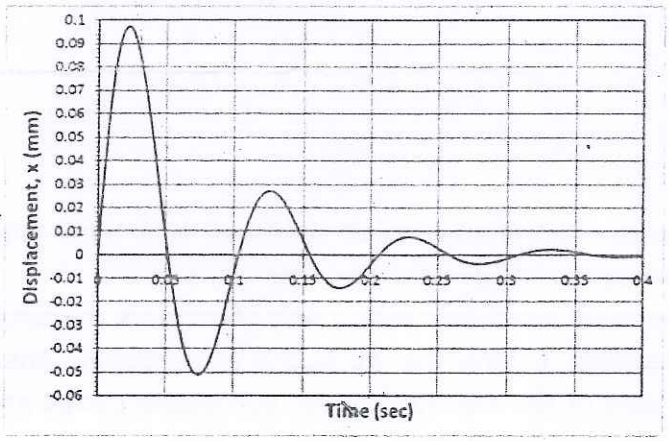
5. a. Describe the rotating unbalance of a machine under forced vibrations and obtain equations for steady state amplitude and phase angle. 07 Marks

b. The wing, engine and the mounting are modelled as a single DOF system with only vertical motion and assuming the wing is fixed. Vibration levels are checked during maintenance by impulse hammer test which is used to supply an initial velocity to the engine. The resulting free vibration of the engine is shown in the figure. From this data, find the approximate values for the damped natural period, viscous damping coefficient, natural frequency and displacement function. 07

(a) Aircraft Engine



(b) Simplified Model



6. a. Explain briefly about Centrifugal pendulum absorber with a neat diagram. **07**

b. A seismic instrument is mounted on a machine running at 800rpm. The natural frequency of seismic instrument is 20 rad/s. The instrument records relative amplitude of 0.8mm. Compute the displacement, velocity and acceleration of the machine. Neglect damping. **07**

7. a. Obtain the natural frequencies of a multi (three) degree of freedom spring mass system by matrix iteration method if K_1=3k, k_2=k_3=k, m_1=4m, m_2=2m, m_3=m. **09**

b. Obtain the natural frequencies of a multi degree of freedom system by Rayleigh's method. **05**

8. Obtain the angular displacement as a function of x and t for the torsional vibrations of circular shaft. **14**